Assignment No : 5

Case Study : Employee Appraisal System

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- **Cursors**

1. Count Employees by Performance Category

```
mysql> DELIMITER $$
mysql>
mysql> CREATE PROCEDURE CountPerformance()
    -> BEGIN
    ->     DECLARE done INT DEFAULT 0;
    ->     DECLARE rating INT;
    ->
    ->     DECLARE excellent INT DEFAULT 0;
    ->     DECLARE good INT DEFAULT 0;
    ->     DECLARE poor INT DEFAULT 0;
    ->
    ->     DECLARE cur CURSOR FOR SELECT Rating FROM appraisal;
    ->
    ->     DECLARE CONTINUE HANDLER FOR NOT FOUND SET done = 1;
    ->
    ->     OPEN cur;
    ->
    ->     loop1: LOOP
    ->         FETCH cur INTO rating;
    ->
    ->         IF done THEN
    ->             LEAVE loop1;
    ->         END IF;
    ->
    ->         IF rating >= 4 THEN
    ->             SET excellent = excellent + 1;
    ->         ELSEIF rating >= 3 THEN
    ->             SET good = good + 1;
    ->         ELSE
    ->             SET poor = poor + 1;
    ->         END IF;
    ->
    ->     END LOOP;
    ->
    ->     CLOSE cur;
    ->
    ->     SELECT excellent AS Excellent, good AS Good, poor AS Poor;
    ->
    -> END$$
Query OK, 0 rows affected (0.06 sec)

mysql>
mysql> DELIMITER ;
```

Testing:

```
mysql> CALL CountPerformance();
+-----------+------+------+
| Excellent | Good | Poor |
+-----------+------+------+
|         0 |    0 |    5 |
+-----------+------+------+
1 row in set (0.00 sec)
```

2. Find Highest Rating Employee

```
mysql> DELIMITER $$
mysql>
mysql> CREATE PROCEDURE HighestRatingEmployee()
    -> BEGIN
    ->     DECLARE done INT DEFAULT 0;
    ->     DECLARE e_id INT;
    ->     DECLARE e_name VARCHAR(100);
    ->     DECLARE rating INT;
    ->
    ->     DECLARE max_rating INT DEFAULT 0;
    ->     DECLARE top_employee VARCHAR(100);
    ->
    ->     DECLARE cur CURSOR FOR
    ->     SELECT e.Emp_ID, e.Name, a.Rating
    ->     FROM employee e
    ->     JOIN appraisal a ON e.Emp_ID = a.Emp_ID;
    ->
    ->     DECLARE CONTINUE HANDLER FOR NOT FOUND SET done = 1;
    ->
    ->     OPEN cur;
    ->
    ->     loop1: LOOP
    ->         FETCH cur INTO e_id, e_name, rating;
    ->
    ->         IF done THEN
    ->             LEAVE loop1;
    ->         END IF;
    ->
    ->         IF rating > max_rating THEN
    ->             SET max_rating = rating;
    ->             SET top_employee = e_name;
    ->         END IF;
    ->
    ->     END LOOP;
    ->
    ->     CLOSE cur;
    ->
    ->     SELECT top_employee AS Top_Employee, max_rating AS Rating
    -> END$$
Query OK, 0 rows affected (0.01 sec)

mysql>
mysql> DELIMITER ;
```

Testing:

```
mysql> CALL HighestRatingEmployee();
+---------------+--------+
| Top_Employee | Rating |
+---------------+--------+
| Amit          |      5 |
+---------------+--------+
1 row in set (0.01 sec)
```

3. Update Low Performers

```
mysql> DELIMITER $$
mysql>
mysql> CREATE PROCEDURE UpdateLowPerformers()
    -> BEGIN
    ->     DECLARE done INT DEFAULT 0;
    ->     DECLARE e_id INT;
    ->     DECLARE rating INT;
    ->
    ->     DECLARE cur CURSOR FOR
    ->         SELECT Emp_ID, Rating FROM appraisal;
    ->
    ->     DECLARE CONTINUE HANDLER FOR NOT FOUND SET done = 1;
    ->
    ->     OPEN cur;
    ->
    ->     loop1: LOOP
    ->         FETCH cur INTO e_id, rating;
    ->
    ->         IF done THEN
    ->             LEAVE loop1;
    ->         END IF;
    ->
    ->         IF rating < 3 THEN
    ->             UPDATE employee
    ->             SET Salary = Salary - 1000
    ->             WHERE Emp_ID = e_id;
    ->         END IF;
    ->
    ->     END LOOP;
    ->
    ->     CLOSE cur;
    ->
    -> END$$
Query OK, 0 rows affected (0.01 sec)

mysql>
mysql> DELIMITER ;
```

Testing:

```
mysql> CALL UpdateLowPerformers();
Query OK, 0 rows affected (0.00 sec)
```

4. Display Employee Appraisal Count:

```
mysql> DELIMITER $$
mysql>
mysql> CREATE PROCEDURE AppraisalCountCursor()
    -> BEGIN
    ->     DECLARE done INT DEFAULT 0;
    ->     DECLARE e_id INT;
    ->     DECLARE total INT;
    ->
    ->     -- Declare the cursor
    ->     DECLARE cur CURSOR FOR
    ->         SELECT Emp_ID, COUNT(*) FROM appraisal GROUP BY Emp_ID;
    ->
    ->     -- Declare handler to exit loop
    ->     DECLARE CONTINUE HANDLER FOR NOT FOUND SET done = 1;
    ->
    ->     OPEN cur;
    ->
    ->     loop1: LOOP
    ->         FETCH cur INTO e_id, total;
    ->
    ->         IF done THEN
    ->             LEAVE loop1;
    ->         END IF;
    ->
    ->         -- Output the result
    ->         SELECT CONCAT('Employee ', e_id, ' has ', total, ' appraisals') AS Result;
    ->
    ->     END LOOP;
    ->
    ->     CLOSE cur;
    -> END$$
Query OK, 0 rows affected (0.01 sec)

mysql>
mysql> DELIMITER ;
```

Testing:

```
mysql> DELIMITER ;
mysql> CALL AppraisalCountCursor();
+----------------------------+
| Result                     |
+----------------------------+
| Employee 1 has 1 appraisals |
+----------------------------+
1 row in set (0.01 sec)

+----------------------------+
| Result                     |
+----------------------------+
| Employee 2 has 1 appraisals |
+----------------------------+
1 row in set (0.01 sec)

+----------------------------+
| Result                     |
+----------------------------+
| Employee 3 has 1 appraisals |
+----------------------------+
1 row in set (0.01 sec)

+----------------------------+
| Result                     |
+----------------------------+
| Employee 4 has 1 appraisals |
+----------------------------+
1 row in set (0.02 sec)

+----------------------------+
| Result                     |
+----------------------------+
| Employee 5 has 1 appraisals |
+----------------------------+
1 row in set (0.03 sec)

Query OK, 0 rows affected (0.04 sec)
```

5. Delete Very Low Ratings

```
mysql>
mysql> CREATE PROCEDURE DeleteLowRatings()
    -> BEGIN
    ->     DECLARE done INT DEFAULT 0;
    ->     DECLARE a_id INT;
    ->     DECLARE rating INT;
    ->
    ->     DECLARE cur CURSOR FOR
    ->         SELECT Appraisal_ID, Rating FROM appraisal;
    ->
    ->     DECLARE CONTINUE HANDLER FOR NOT FOUND SET done = 1;
    ->
    ->     OPEN cur;
    ->
    ->     loop1: LOOP
    ->         FETCH cur INTO a_id, rating;
    ->
    ->         IF done THEN
    ->             LEAVE loop1;
    ->         END IF;
    ->
    ->         IF rating = 1 THEN
    ->             DELETE FROM appraisal
    ->             WHERE Appraisal_ID = a_id;
    ->         END IF;
    ->
    ->     END LOOP;
    ->
    ->     CLOSE cur;
    ->
    -> END$$
Query OK, 0 rows affected (0.01 sec)
```

Testing:

```
mysql> CALL DeleteLowRatings();
Query OK, 0 rows affected (0.00 sec)
```